Chapter 12, Problem 1PP

Problem

Given that $\mathbf{V}_{bn} = 220\angle 30°\ \text{V}$ find **V***an* and **V***cn*, assuming a positive (*abc*) sequence.

Step-by-step solution

Step 1 of 3

Refer to Figure 12.7 in the textbook.

Consider the value of $\mathbf{V}_{bn}$.

$$\mathbf{V}_{bn} = 220\angle 30°\ \text{V}$$

Compare with $V_p\angle\theta°$.

Therefore, the value of V_p is $220\ \text{V}$ and $\theta°$ is $30°$.

Step 2 of 3

Notice that $\mathbf{V}_{an}$ leads V_{bn} by $120°$.

Calculate the value of $\mathbf{V}_{an}$.

$$\mathbf{V}_{an} = V_p\angle(\theta° + 120°)$$

Substitute $220\ \text{V}$ for V_p and $30°$ for $\theta°$.

$$\begin{aligned}\mathbf{V}_{an} &= 220\angle(30° + 120°)\\ &= 220\angle 150°\ \text{V}\end{aligned}$$

Hence, the value of the phase voltage $\mathbf{V}_{an}$ is $\boxed{220\angle 150°\,\text{V}}$.

Step 3 of 3

Notice that $\mathbf{V}_{bn}$ leads $\mathbf{V}_{cn}$ by $120°$.

Calculate the value of $\mathbf{V}_{cn}$.

$$\mathbf{V}_{cn} = V_p\angle(\theta° - 120°)$$

Substitute $220\ \text{V}$ for V_p and $30°$ for $\theta°$.

$$\begin{aligned}\mathbf{V}_{cn} &= 220\angle(30° - 120°)\\ &= 220\angle -90°\ \text{V}\end{aligned}$$

Hence, the value of the phase voltage $\mathbf{V}_{cn}$ is $\boxed{220\angle -90°\,\text{V}}$.